



ABDERA TRADING

Quantitative Research

What the work looks like, told through one real day of it

This pack walks through one complete day of quantitative research: a hypothesis taken from the literature to a measured verdict, and a deliverable shipped.

The headline idea died. The thing that shipped came from somewhere else.

That is the ordinary shape of the work, which is why this is the pack we send rather than a success story.

Companion code, figures and a candidate exercise accompany this document.

ABD-ONB-R1

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1. The role, and what this pack is

You are looking at the complete record of one day of quantitative research at Abdera Trading: one hypothesis taken from the literature to a measured verdict, and one deliverable shipped. Nothing here is idealised. The headline idea **died**, and the thing that shipped came from somewhere else. That is the ordinary shape of the work, and it is why this is the pack we send to candidates rather than a success story.

If you are strong in mathematics, statistics or physics and have never done this before, this pack is written for you. It assumes you can read a derivation and write clean code. It assumes nothing about markets.

What the job actually is

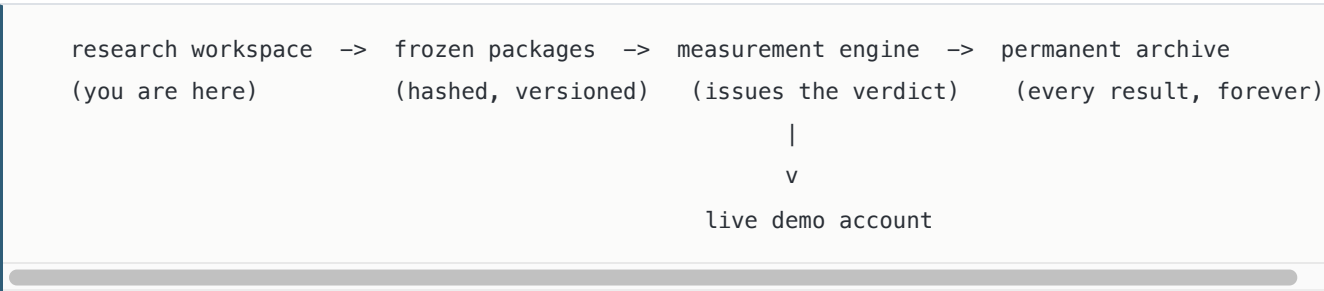
A quant researcher here turns a vague idea into a falsifiable claim, tests it so that the answer can be trusted, and writes down what was learned either way.

Notice what is not in that sentence. Not "finds strategies that make money" — that is the outcome we want, but if you optimise for it directly you will produce results that look profitable and are not, which is worse than producing nothing. The discipline is the product. A well-evidenced negative result is a success, and we treat it as one.

The realistic ratio, from Abdera's own record: across research batches 005 to 016, more than twenty families were tested. Almost all were rejected. Two survived far enough to be measured at tick level by the firm's engine. One of those runs on a live demo account today.

If a month of careful work ending in "this does not work, here is the number that closed it" would feel like failure to you, this is the wrong job. If it feels like a result, read on.

The five stages



Stage	The question it answers
Research workspace	Is this idea worth an hour of tick-level measurement?
Frozen packages	Is this package hashed, reproducible, and exactly what was measured?
Measurement engine	Does it survive the firm's risk model, and can the result be told from luck?
Archive	What did we try, what happened, where is the evidence?
Live demo	Which ones get real money, and what does the operator need to know?

You work in stage one. Almost everything in this pack is about stage one, because that is the job being offered.

A result produced in the research workspace is a screening decision, not a measurement. When we say a candidate is worth promoting, we mean "worth an hour of tick-level measurement downstream" — never "this works". Every dollar figure a researcher produces is pre-measurement. The word *verdict* belongs to the measurement engine alone, and it is a different team's word.

That distinction is not modesty. The research engine simulates fills on one-minute bars; the measurement engine walks every tick against the real bid and ask under the firm's risk model. A matching signal trace is not a matching profit path.

Where you start: a \$50,000 prop account, not the firm's capital

New researchers do not trade the firm's own capital. You start on **proprietary-trading-firm evaluation accounts, at the \$50,000 size**, and everything in this pack is written for that setting.

This is deliberate, and it is worth understanding before you decide whether the job appeals.

It bounds the downside while you build a track record. A prop account has a hard, external drawdown limit. The worst case is a failed evaluation and a lost subscription fee — a known, small, bounded number. Nobody has to decide how much of the firm's balance sheet to hand a researcher whose work nobody has seen yet, which means you get to start immediately rather than after months of proving yourself on paper.

It is real money under real rules. This is not a simulator. The account is live, the drawdown is enforced by someone else, the daily flatten is mandatory, and the fee is charged whether or not you pass. Every constraint below is a real constraint, and a strategy that ignores one does not merely score badly — it fails.

The constraints are what make the problem specific. A researcher told "make money in futures" has an unbounded, unfalsifiable brief. A researcher told "pass a \$3,000 target against a \$2,000 ratcheting drawdown, flat by the close, while paying monthly rent" has a real optimisation problem with a measurable objective. Most of the interesting results in this pack — including the counterintuitive one in section 9, where *more* risk per trade turned out to be correct — exist only because the constraints are sharp.

Progression follows a track record, not a job title. Researchers whose candidates survive tick-level measurement and hold up on live demo accounts take on more, including allocations of the firm's own capital. That path is earned with measured results, and the prop route is where those results get made. Nobody here started anywhere else.

What we trade, and the constraint that shapes everything

Nasdaq and S&P futures, intraday, inside those evaluation accounts. The account rules are not a detail — they are the objective function:

- a **\$50,000** notional account with a **\$2,000** trailing drawdown that ratchets up and never retreats;
- a **\$3,000** profit target to pass;
- a **mandatory flatten** every afternoon, so nothing is held overnight;
- a **monthly subscription** for as long as the attempt takes.

That last one changes the mathematics of the whole problem. Because the account pays rent by the month, **time is a cost**. A strategy that passes more reliably but takes three times as long can be worth less than one that fails more often and resolves quickly. This is not intuition; it is measured, and it is in section 9.

How to read this pack

docs/01	Where ideas come from, and the filter every idea passes before it costs anything
docs/02	The mathematics you do before writing code
docs/03	The data we hold, its traps, and how you get data we do not hold
docs/04	Building the first version, and the tests that catch a broken one
docs/05	Scaling the test: preregistration, controls, kill gates
docs/06	Promotion, packaging, and how a strategy gets its identity
docs/07	Worked example one: the idea that died. Read this one closely
docs/08	Worked example two: the thing that shipped
docs/09	The catalogue of what gets work thrown out
code/	Runnable implementations of everything above
exercise/	A task. If you want to be taken seriously as a candidate, do it

Everything in `code/` runs with Python 3 and numpy and nothing else. Start with:

```
python3 code/preflight.py
```

2. Finding an idea

Where they come from

Four sources, in rough order of how often they produce something worth testing.

Academic literature. SSRN, arXiv q-fin, the Journal of Financial Economics, the Review of Financial Studies. Papers on market microstructure, dealer intermediation, option hedging flows, and order-flow imbalance are the productive neighbourhoods for intraday index futures. A published result is not a strategy — sample periods end, effects decay, and transaction costs are usually understated or absent — but a published result comes with a *mechanism*, and the mechanism is what you are shopping for.

Practitioner writing. Desk commentary, options-flow services, market-structure blogs. Lower signal-to-noise than the literature and rarely quantified, but this is where you learn what participants actually believe and act on, which is itself tradeable information. The worked example in section 8 came from here.

Exchange and regulator documentation. Contract specifications, settlement procedures, margin rules, expiration mechanics. Dry, and it is where the genuinely structural effects live, because they come from rules rather than from behaviour.

Anomalies you observe. Something in the data that you cannot explain. Treat these with the most suspicion — you found them by looking, which means the multiple-testing problem starts before you have written anything down.

The filter every idea passes first

Before an idea costs anything, it has to survive two questions.

Who pays, and why?

Name the counterparty who loses the money you expect to make, and the reason they are willing to. Acceptable answers look like:

- *a dealer who must hedge an option book continuously, whose flow is mechanical and price-insensitive, and who is paying for immediacy;*
- *an index fund that must trade at the close regardless of price;*
- *a leveraged participant forced to liquidate by a margin rule.*

Unacceptable answers: "the market is inefficient", "the pattern is visible in the chart", "momentum exists". If you cannot name the payer, you do not have a hypothesis — you have a shape.

Has it already been closed?

The firm keeps a register of falsified hypotheses. Each entry records what was tested, on what data and at what horizon, what closed it, and **what would have to change for a retest to be legitimate**. Read it before proposing anything.

This matters more than it sounds. A family being falsified at one horizon is not falsified at all of them, and a family being popular is evidence of neither. In the worked example, what was closed was *direction*, from the *overnight* option book, at *session* scale, on the S&P contract. Intraday-updated positioning is a different

object and remains untested. Saying which one you mean is the difference between a new hypothesis and a repeat.

Writing the hypothesis down

A usable hypothesis names five things. If you cannot fill these in, you are not ready to write code.

Mechanism	who is forced to trade, and why
Information set	exactly which fields, known at exactly what time
Instrument and horizon	what you trade, and over what holding period
Prediction	the direction and rough size of the effect, stated before you look
Falsifier	the observation that would make you abandon it

Here is the one from the worked example, written before any data was pulled:

Mechanism. Option dealers delta-hedge continuously. When their book is net short gamma they must trade in the direction of the move, amplifying it; when net long gamma they trade against it, damping it. The hedging flow is mechanical, price-insensitive, and its *location* is computable in advance from the option chain. **Information set.** Open interest by strike and expiry, published overnight and describing the prior close; option greeks and implied volatility from the prior session's end-of-day report. Both known before the opening auction. **Instrument and horizon.** S&P futures, one session, entry after the open and exit before the mandatory flatten. **Prediction.** Conditional on the regime, price behaves differently near the computed "flip" level than near an arbitrary level: continuation in short gamma, reversion in long gamma. **Falsifier.** The difference between regimes is indistinguishable from zero, or the real level is indistinguishable from a level a hundred ticks away.

Note the last line. **The falsifier was written before the data existed**, and it is what the test was eventually designed to hit. It hit.

What it costs to be wrong later rather than early

The pre-flight arithmetic in the next section takes ten minutes. Implementation takes days, and a full test takes a week of wall-clock time. The ordering is not a formality: an idea killed by arithmetic costs ten minutes, and the identical idea killed after implementation costs a week and leaves you invested in it, which is the more expensive of the two problems.

3. The mathematical framework

This section is the one to read carefully. Everything here is done **before** any code is written, and most ideas die in it.

Runnable: `python3 code/preflight.py`, `python3 code/gex_math.py`, `python3 code/gates.py`.

3.1 Barrier geometry, or why most ideas are dead on arrival

Almost every intraday strategy here is a **bracket**: enter, and exit at whichever of two barriers is touched first — a stop S ticks against you, or a target T ticks in your favour.

Model the price path as a driftless random walk. The probability of touching the target before the stop is the classic gambler's-ruin result, and it depends only on the distances:

$$p_{\text{null}} = \frac{S}{S+T} = \frac{1}{1+RR}, \quad RR \equiv \frac{T}{S}$$

This is the win rate your strategy gets **for free, with no edge at all**. It is the number any observed win rate must be compared against, and it is why a high win rate on its own is not evidence of anything: widen the stop and shrink the target and you can manufacture a 79% win rate out of pure noise, with an expected profit of exactly zero.

Now add cost. Let C be the round-turn transaction cost in ticks. Breakeven requires

$$pT - (1-p)S = C \quad \implies \quad p^* = \frac{S+C}{S+T}$$

Subtract the free win rate, and everything collapses to one expression:

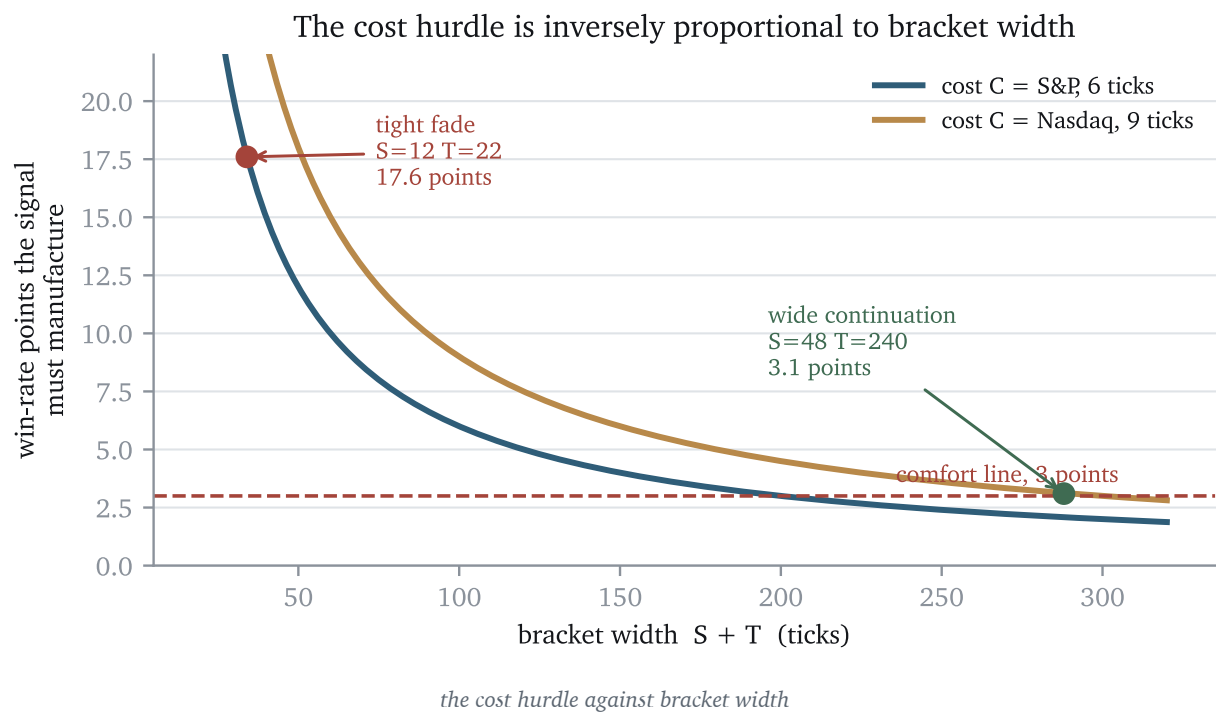
$$p^* - p_{\text{null}} = \frac{C}{S+T}$$

The entire cost hurdle, expressed in win-rate percentage points, is inversely proportional to the total width of the bracket. This is the single most useful formula in the pack.

Read what it implies. A tight mean-reversion bracket of $S = 12$, $T = 22$ ticks against a cost of 6 ticks needs the signal to manufacture $6/34 = 17.6$ percentage points of win rate over the driftless null. No equity-index signal delivers that. A wide continuation bracket of $S = 48$, $T = 240$ against a cost of 9 ticks needs $9/288 = 3.1$ points. Same instrument, same cost, one idea impossible and the other worth building — and you know which before writing a line of code.

We require a margin over breakeven, not breakeven itself, so the operative threshold is

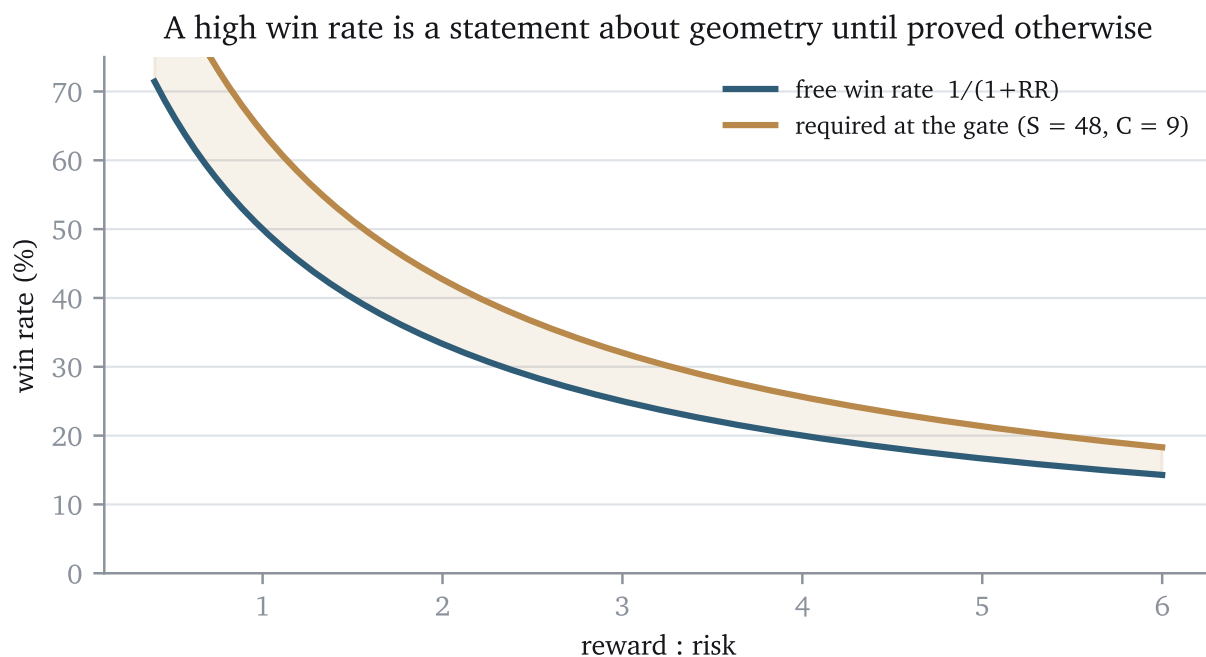
$$p_{\text{gate}}^* = \frac{S+\lambda C}{S+T}, \quad \lambda = 1.5$$



The rule of thumb

Cost burden $C/(S+T)$	Reading
under ~3 points	the geometry is affordable; the signal only has to be real
3 to 6 points	you need a named mechanism for the excess
above ~6 points	the geometry is wrong; fix it or abandon the idea

And the rule that follows: **nothing downstream of the entry decision can move mean gross ticks per contract**. Position sizing, risk management, account selection and lockout logic move dollars around. They do not move ticks. Ticks are a property of the signal and the barrier alone, and that is the number that has to be right first.



the free win rate, and what the gate demands

3.2 Why R-multiples, not ticks, when stops vary

If a strategy clips its stop into a range — say 40 to 120 ticks — then pooling raw tick outcomes across trades is a mistake, and an expensive one.

Under a bracket the outcome is $+RR \cdot S$ or $-S$, so

$$\text{Var}(\text{gross} \mid S) = S^2 p(1 - p) (1 + RR)^2$$

Variance scales with S^2 . Across a 3-to-1 spread of stops that is a **9-fold variance ratio** between the widest and tightest trades, so the wide-stop trades set the standard error almost by themselves. The resulting t-statistic answers "did the wide-stop trades win", not "does the signal have edge".

Report the **R-multiple** instead: for each trade,

$$R_i = \frac{\text{gross ticks}_i / q_i - C}{S_i}$$

with its mean, its session-clustered standard error, and the t-statistic. In this firm's history a strategy showed **+6.75** net ticks at **t = 1.25** in raw ticks — a marginal but live-looking result — and **+0.043 R** at **t = 0.57** once expressed properly, which also revealed that the tight-stop half was losing money after costs the whole time.

3.3 The specific model: dealer gamma

The worked example needs a model that turns an option chain into a price level. This is what it looks like when a hypothesis is made concrete. Full implementation: `code/gex_math.py`.

Gamma is the second derivative of option value with respect to spot — the rate at which a hedger's delta changes as price moves:

$$\Gamma = e^{-q\tau} \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}} \quad d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}$$

where S spot, K strike, sigma implied volatility, tau time to expiry, r rate, q dividend yield, phi the standard normal density.

Gamma is **identical for a call and a put at the same strike and expiry**. That fact is load-bearing and we return to it.

Convert to the dollar hedge a 1% move forces, per contract line:

$$DG_i = \Gamma_i \cdot OI_i \cdot m \cdot S^2 \cdot 0.01$$

where **OI** is open interest and **m** the contract multiplier. One factor of **S** converts gamma into delta-per-1%-move; the second converts delta into dollars.

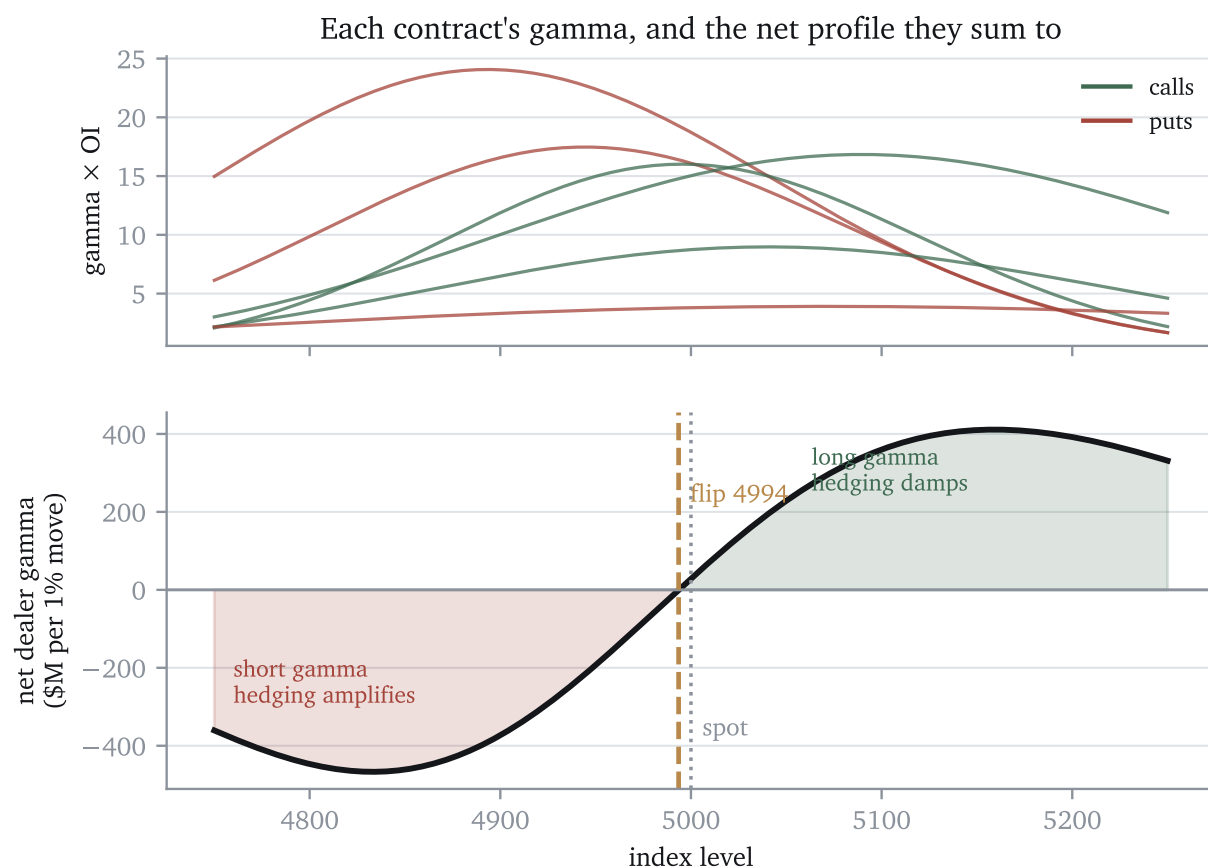
Aggregate with a sign for which side the dealer is on:

$$N(S) = \sum_i \epsilon_i DG_i(S), \quad \epsilon_i \in \{-1, +1\}$$

and define the **flip level** as the zero crossing of **N** nearest spot:

$$N(S^*) = 0$$

Above it dealers are net long gamma and hedge against the move; below it they are net short and hedge with it. That is the theory being tested.



per-contract gamma, the net profile, and the flip level

The assumption hiding in plain sight

Open interest counts contracts. **It does not say who is long and who is short.** The signs ϵ_i are a *convention*, not data — the standard one assigns dealers long calls and short puts — and since gamma is identical for calls and puts, that convention is the model. Everything the hypothesis claims rests on it.

So it gets isolated in one function and tested against its own inversion and against random signs. This produces a result worth internalising, and `code/gex_math.py` verifies it numerically: inverting the convention negates $N(S)$ exactly, so the flip **location** is convention-invariant while the **regime label** flips. A test whose result survives the inversion unchanged has learned nothing about dealer positioning.

The general principle: find the modelling assumption that your result depends on, isolate it in one place, and make its ablation a first-class output. Every model has one. If you cannot name yours, you have not found it yet.

3.4 Getting from an index level to a tradeable price

The option chain prices the cash index; you trade a futures contract on a discrete tick grid. They differ by **basis** — carry, dividends and financing:

$$b = F_{\text{close}} - I_{\text{close}}, \quad \text{tick index} = \text{round}\left(\frac{L + b}{\delta}\right)$$

where F futures close, I index close, L the level in index points, δ the tick size.

measured from the same instant on both sides, never assumed. Two rules here are absolute:

1. **Every price the engine hands a strategy is an integer tick index, not a price.** Mixing the two units is silent — the comparison does not raise, it produces a number — and it has destroyed three strategies in this firm's history. Section 10.
2. Measure the basis; do not model it. A measured constant per session is fine. A theoretical cost-of-carry is a second model stacked on the first.

3.5 Deciding whether a difference is real

Implementation: `code/gates.py`.

Sessions are the unit of independence

Trades within one session share a regime, a news cycle and an opening auction. Resampling *trades* understates uncertainty; we resample whole **sessions**. The variance inflation from clustering is

$$\text{design effect} = \sqrt{1 + (m - 1)\rho}$$

where m trades per session, ρ the within-session correlation.

for m trades per session and within-session correlation ρ . At one trade per session it is negligible. At five trades and $\rho = 0.3$ it is **1.48**, which is the difference between $t = 2.1$ and $t = 1.4$ — between a result and nothing.

For a difference between two groups of sessions, resample the sessions in each group independently and take the 2.5th and 97.5th percentiles of the difference. Nothing more exotic is needed, and nothing less is honest.

A maximum over many cells is not a discovery

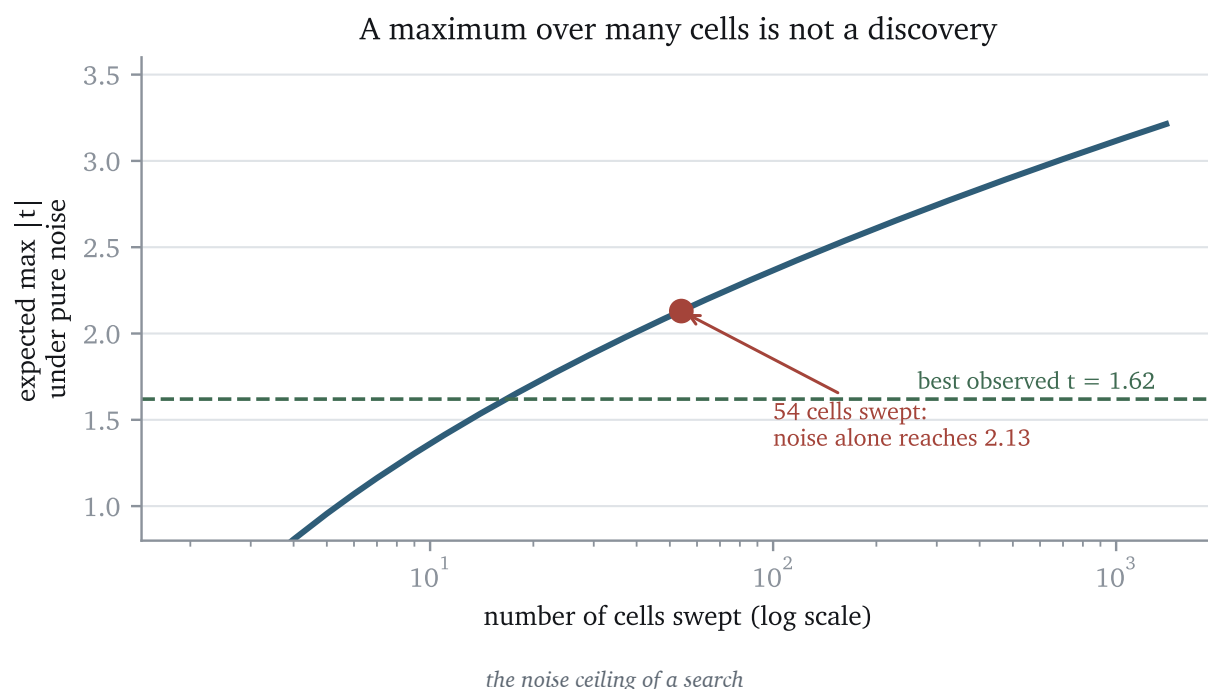
If you sweep n variants, the best one always looks good. The honest comparison is against what the best of pure noise would have produced. For n independent standard normal draws,

$$\mathbb{E}[\max] \approx \sqrt{2 \ln n} - \frac{\ln \ln n + \ln 4\pi}{2\sqrt{2 \ln n}}$$

cells swept	expected max \	t\	under pure noise
1	0.00		
20	1.71		
54	2.13		
200	2.61		
1,287	3.19		

A real sweep from the workspace: 54 cells of vol-surface features against next-session direction. Best t-statistic in the exploration window: **1.62**. Independent noise alone would be expected to reach 2.13. **The**

search did not clear its own null — as clean a negative as this method produces, and it took one line of arithmetic to see it.



Correlated cells lower the ceiling, so treat this as the generous version of the comparison.

Controls: make the null concrete

An interval that excludes zero says the difference is not noise. It does not say the difference is caused by what you think. That needs controls, and each is cheap:

Control	What it kills
Phantom levels — repeat everything at the level $\pm$ 100/150/200 ticks	the level is arbitrary; any nearby level would have done
Sign / label scramble — randomise the modelling assumption	the classification carries no information
Timing null — draw a random minute in the same session	the day-selection carries it, the entry adds nothing
Side null — take the same trades with random direction	the direction carries nothing

The phantom control is the one candidates most often omit and the one that does the most work. In the worked example, the real level's effect exceeded the phantom pool's by -5.16 ticks with a 95% interval of $[-19.70, +10.55]$. The level was not special, and no amount of interval-tightening on the primary endpoint would have revealed that.

Explore and confirm

Split the history once, in advance. Iterate freely in the exploration window; look at the confirmation window once, at the end, and report both columns side by side for every cell you show.

A cell that is strong in exploration and inverts in confirmation is not a weak result — it is a **negative** result, and reporting it as "promising, needs more data" is the most common way a research programme lies to

itself. In the sweep above, the best exploration cell went from `+29.0` ticks at `t = 2.6` to `-6.1`, and the theory-aligned cell went from `+31.1` to `-32.9`.

3.6 The pre-flight, as a checklist

Every line is a kill switch. `code/preflight.py` runs it.

#	Quantity	Kill if
1	round-turn cost <code>C</code> , stress column	—
2	required gross edge <code>1.5C</code>	your honest estimate of mean gross is below it
3	reward:risk <code>RR = T/S</code>	<code>RR < 1</code> and no mechanism gives a 65%+ hit rate
4	geometric breakeven <code>1/(1+RR)</code>	—
5	win-rate excess <code>C/(S+T)</code>	above ~ 3 points with no named mechanism
6	required win rate <code>(S+1.5C)/(S+T)</code>	above 0.60
7	expected trade count	below 100 over the intended window
8	stop dispersion <code>max(S)/min(S)</code>	above ~ 2 and you are not reporting R
9	does any feature cross a session boundary?	yes, and you have not handled the contract roll
10	does sizing read a ratcheting quantity?	yes

4. Data

What the firm holds

The futures tick dataset

The primary source. Everything that touches price comes from it.

Size	2.74 billion trades, 6.6 GB
Instruments	S&P and Nasdaq futures, three FX pairs, two crypto contracts
Span	2020-08 to 2026-08
Per trade	price, size, the best bid and ask at the moment of the trade , and the aggressor side
Order book	top ten levels on a one-second grid, S&P and Nasdaq, from 2025-10 only
Prices	integer tick counts , not floats

The bid, ask and aggressor on every trade matter more than the size of the file. They mean you can *measure* your cost assumption rather than inherit one, and that a fill model can be checked against what actually traded.

The options dataset

Added mid-2026 for the worked example in section 8. End-of-day greeks, implied volatility and open interest per contract per day, for the S&P index option complex and the two large index ETFs; 1,541 sessions per root, aligned to the futures calendar.

What we do not have

No history before 2020-08 — no COVID crash, no 2018 volatility regime. This is the hardest constraint on what any result here can claim. Six years is enough for an honest out-of-sample split; it is not enough to claim a rule survives a regime it has never seen, and saying so is part of the job.

No queue position and no order lifetime. The book is one-second snapshots, not an event stream, so strategies that depend on queue priority cannot be tested honestly and are deferred rather than approximated.

If you need data we do not have, we buy it

This is a real policy and it is worth stating plainly, because it is unusual and it is one of the better reasons to work here.

If an idea is worth testing and the data to test it does not exist in-house, the firm acquires it. Any commercial API, any vendor feed, any historical archive. You do not have to make a business case for a subscription before you are allowed to have a hypothesis.

The worked example in section 8 is exactly this. The dealer-gamma hypothesis required option greeks and open interest by strike, which the firm did not hold. The subscription was bought **the same day the hypothesis was framed**, the full six-year history was downloaded within the hour, and the hypothesis was tested to a verdict inside two days.

The verdict was that the idea does not work. The subscription was still the right call: the alternative was to keep wondering, and wondering is more expensive than eighty dollars.

What the firm asks in return is only that you know *what you would do with the data* before asking for it — the information set, the timestamps, the fallback if a field turns out to be unavailable. That is section 2's hypothesis template, and it is the whole approval process.

Five traps, all of which have cost this firm something

1. Prices are integer tick counts. Real price is `count × tick size`. This is the correct unit for a strategy and the wrong unit for a human, so convert only at the moment you print something. The failure mode is comparing a converted price against an unconverted indicator — see trap 5.

2. Large trades arrive split into pieces. When one aggressive order fills against several resting orders, the feed emits one record per fill, flagged first / middle / last. To recover exchange-level trades, group each run from a first-flag to the next last-flag and sum the sizes. Compute a trade-size distribution without doing this and you will systematically understate large trades — which are the ones an order-flow hypothesis is about.

3. The sub-millisecond timestamp is a counter, not a clock. It saturates, so in bursts several records share a timestamp. Row order is authoritative. Sorting by timestamp destroys sequence.

4. The trade price is not always inside the recorded quote. Between 1% and 13% of trades print outside the recorded bid-ask, depending on instrument. That is a property of how the quote was captured, not an error. A fill model that asserts otherwise silently rejects real trades.

5. Contract rolls contaminate anything that differences across a session boundary. A futures series is spliced from consecutive delivery months. On a roll day, "today's open minus yesterday's close" is the price difference between two *different instruments* — carry, not information. Measured on the Nasdaq contract: 27 roll sessions in 1,790, with gaps up to **1,131 ticks** against a typical volatility of 10 to 40 ticks.

The consequence was not hypothetical. A filter gated on overnight displacement fired on 70% of roll days against 45% of ordinary days, and because carry was positive on every roll after 2023 it fired **systematically long** — 16 of 19 times. Handle it one of two ways, back-adjust the series or refuse roll sessions entirely, and **state in your notes which one you chose**.

For any feature conditioned on a gap, report the **roll-day firing ratio**:

$$\text{ratio} = \frac{\text{firing rate on roll days}}{\text{firing rate on other days}}$$

Above roughly 1.5, the feature is reading carry rather than information.

Provenance is part of the result

Two rules that look bureaucratic and are not.

Every dataset carries a manifest with a hash of every file, verified after any move or copy. A result you cannot tie to a specific, verified input is not reproducible, and the firm has been bitten: a legacy archive was deleted, and every result measured on it became a *historical statement* — citable from frozen evidence, not re-derivable. Those results are still in the record, labelled exactly that way.

Vendor semantics are part of the tested claim, so measure them. In the worked example the option open-interest reports were assumed to publish at 06:30 each morning. A publication-time assertion in the feature builder refused two full years of data, which is how we discovered the vendor batch-stamps the early era at 07:01 and emits late tails as far as 20:15. The fix was row-level enforcement of the actual report time, with the excluded mass recorded per session — not a looser clock.

An unverified "this is available at 09:00" line in a protocol is a look-ahead bug waiting to be discovered by someone downstream, and it will be discovered after you have built on it.

5. Building it, and the first small tests

The strategy contract

A strategy is **one file** that runs byte-for-byte in the research workspace, in the firm's measurement engine, and in the live runtime. Annotated template: `code/strategy_template.py`.

The rules that a linter enforces, each of which exists because breaking it destroyed a result:

Rule	Why
One registered strategy class per file, no cross-strategy imports	two classes in one file defeat the identity hashing
Imports from the portable kit only, never engine internals	that is what makes the file portable
No comments and no docstrings	explanation goes in the notes; names carry the meaning
No numeric literals outside declared parameters	every tunable is part of the frozen record
Publish the series you intend to compare, read it back by name	prevents the unit bug
Guard every comparison against an indicator with a finiteness check	<code>nan > x</code> is <code>False</code> , so a missing guard yields a strategy that silently never trades
A signal formed on a bar fills on the next bar or later	anything else is look-ahead

That third rule surprises people. In a research repository, a comment is a claim about the code that no test enforces and that drifts silently away from what the code does. Explanation belongs in the notes file, which is reviewed as documentation; the code should be readable from its names alone. This pack's `code/` directory is annotated because it is teaching material — the real thing is not.

Pilot before you build

Pull a small slice of the data first — a few weeks — and check the shape of it before the hypothesis has any chance to make you credulous.

In the worked example the pilot was three weeks: 76 files, 703,000 rows, one minute of wall clock. It immediately exposed a definition problem that would have silently poisoned everything.

The model computes "walls" — the strikes carrying the most dealer gamma. Computed across the full 90-day option book, the biggest strikes were far-out-of-the-money hedges sitting hundreds of points from the market, which is a true statement about the book and useless for an intraday level. The definition was narrowed to contracts expiring within ten days **before any outcome was opened**, and the change was recorded as a preregistered definition rather than a tuning decision.

The distinction is everything. Narrowing a definition *before* seeing outcomes is modelling. Narrowing it *after* is fitting, and it produces a result that will not survive contact with new data.

The three tests that catch a broken strategy

Run all three before the first real backtest.

Units

Every comparison must be tick index against tick index. Search the file for any call that returns a display price and justify each one. On the Nasdaq contract the two units differ by a factor of four, and the comparison does not raise — it produces a number.

Branch reachability

For every clause of your predicate, count how many bars satisfy it in isolation and cumulatively. **A clause firing on 0% or 100% of bars is a bug until proved otherwise.** This takes ten minutes.

The real output from the shipped strategy in section 9, over 492,417 candidate bars:

clause, applied in order	bars surviving
regular trading hours	492,417
all features finite	446,907
inside the entry window	189,000
overnight displacement gate	67,500
directional drive gate	32,400
not a contract-roll session	32,400
breakout cross, either direction	828

and in isolation, the check that matters most: **4,645 upward crossings against 4,219 downward.** Both directions alive and balanced.

Three strategies in this firm's history failed exactly this. In one, a comparison between a display price and a tick index made the long branch unsatisfiable and the short condition true on *every* bar — the whole branch was dead in both directions, not biased, dead. The strategy ran, reported, and produced a verdict. In another, found during the export in section 9, the same unit error pinned the direction to short on every session and rendered a magnitude threshold inert; it had already been shipped and was waiting to be measured.

Causality

Nothing that computes indicators over the whole history may use information from after the bar it labels. No centring on the full sample, no filters that run forwards and backwards, no fitting on data that includes the future. The engine enforces the obvious cases by refusing negative lags; the subtle ones are yours to avoid, and the standard test is to corrupt the tail of the input and assert that the earlier output is byte-identical.

Then, and only then, the first backtest

One contiguous run over the full history on a single account. Not a grid of overlapping windows — that comes later and answers a different question. Two statistics are mandatory:

- **coverage**: sessions where an order was submitted, divided by eligible sessions;
- **last trade date**, compared against the end of the window.

A run whose last trade is years before the window ends did not measure a signal. It measured a latch. Three strategies at this firm reported healthy status while trading nothing for years, because nothing had breached — because nothing had traded. Section 10 has the causes.

6. Scaling the test

The small tests told you the strategy is not broken. They said nothing about whether it works. This section is how you find out in a way that will still be true next year.

Preregistration

Before any outcome is computed, write the protocol down and freeze it. Its hash goes into the research ledger, and any later change is an **amendment** that records what changed and why — never an edit.

This is not ceremony. Without it, the sequence "compute a result, dislike it, adjust a threshold, recompute" is indistinguishable from research, including to the person doing it. The frozen protocol is what makes the difference visible afterwards.

A protocol states:

Mechanism and payer	why the effect should exist
Information set	every field, with the exact time it becomes knowable
Feature definitions	the formulas, frozen with the implementation version
Partitions	which data may be looked at when
Primary endpoint	one statistic, named before it is computed
Secondary diagnostics	named, and never promotable on their own
Controls	the nulls the effect must beat
Kill gates	in order, with thresholds, and the stop-on-failure rule
Known limitations	stated up front, not discovered in review

One primary endpoint. Not a family of related statistics from which the best is reported. The arithmetic in section 3.5 is why: with 54 cells, pure noise reaches $t = 2.1$, and `code/gates.py` demonstrates the same thing empirically — a 95% interval is wrong one time in twenty by construction, so a single significant cell out of many is the *expected* behaviour of the method, not a discovery.

Partitions

Split the history once, in advance, and respect the split:

Partition	Rule
Development	open. Iterate freely here
Calibration	sealed. Opened once per frozen candidate; no retuning afterwards
Forward	candidate-blind. Single shot, and only on explicit authorisation

Data quality checks — coverage, missingness, distributions — are not outcomes and may run on any partition at any time. Anything that joins your feature to a subsequent price move is an outcome and obeys

the table.

Kill gates

Ordered, thresholded, and evaluated in sequence. **Stop at the first failure.**

A representative set, from the worked example:

1. **Sample.** Fewer than 100 events per group → dead.
2. **Primary differential.** Its 95% session-bootstrap interval includes zero → dead.
3. **Beats phantom levels.** The real level's effect does not exceed levels 100 to 200 ticks away → dead; the level is not special.
4. **Implementation consistency.** The modelling assumption's inversion produces the exact arithmetic negation it must. A deviation is a bug, not a finding.
5. **Survives the scramble.** A significant effect with randomised labels → dead; the labelling carries nothing.
6. **Era agreement.** Sign disagreement across the sample's structural break → flagged, and any surviving claim must be stated as the narrower claim.

Stopping matters more than the thresholds. Once a gate has failed, every further statistic is a search for a reason to continue, and a programme that always finds one never kills anything. When the worked example failed gate 2, gates 3 to 6 were **not evaluated** — and an unevaluated gate is never a pass. `code/gates.py` implements exactly this and prints what it did not evaluate.

What a promotable result looks like

Six things, all of which are properties of the evidence rather than of the strategy:

- **A contiguous single-account run** over the full history, reporting coverage and last trade date. A grid of overlapping windows does not substitute: each window restarts a fresh account, so the design is *structurally blind* to any state that survives a day — a latch, a counter, an account floor. Two strategies here scored well on such a grid while trading almost nothing in a continuous account. Both reports were arithmetically correct about different experiments; only one was about the strategy.
- **quantity == 0 recorded as an event.** A strategy that declines to size is otherwise indistinguishable in the output from one that saw no signal.
- **Every latch documented with its clearing condition, and a test for it.** Initialisation at start-up is not a clearing condition.
- **Time-to-pass and accumulated fees**, not only profit. For an account that pays monthly rent, the median days to clear the target and the fees spent getting there decide the economics more than the equity curve does.
- **Both account plans swept.** The same signal on the same tape passed at 30.4% on one plan and 12.6% on another, entirely because of the firms' rules. A strategy whose edge concentrates in one trade is structurally incompatible with a consistency rule, and that is knowable before any backtest.
- **Concentration.** The mean after removing the top 5% of trades, beside the untrimmed mean, and a per-year panel. A positive mean that turns negative after trimming is a lottery ticket, and this has killed more candidates here than bad signals have.

Two diagnostics worth more than they look

A stable pass rate with collapsing profit. One strategy held a 30.4% pass rate across an out-of-sample cut while net profit went from $+\$1,400$ to $-\$16,106$. Cadence durable, payoff not: the winning trades got smaller, not rarer. That single comparison says where to work — on what happens *after* entry, not on when to enter.

A better pass rate that loses money. Reducing risk per trade raised a strategy's pass rate from 42.9% to 66.7% and dropped net profit from $+\$13,989$ to $+\$148$. Smaller size meant twice as many trades per attempt, each attempt ran far longer, and each accumulated more monthly rent. Higher pass rate, fewer passes, same rent, less money.

Both are counterintuitive, both are measured, and neither is visible if you report profit alone.

7. Promotion, packaging and identity

What leaves the workspace

When a candidate survives, it is **frozen** into a package and handed to the measurement team. The package is four files:

File	Contents
<code>strategy.py</code>	the frozen source, byte-for-byte, LF line endings
<code>params.yaml</code>	exactly the class defaults, nothing else
<code>manifest.json</code>	identity, hashes, taxonomy, status, and what could not be verified
<code>NOTES.md</code>	the evidence, in a fixed section order ending in <i>Not verified</i>

The measurement engine reads the **class defaults** and applies overrides on top, so `params.yaml` is documentation rather than configuration. Keeping it equal to the defaults is enforced mechanically: if it ever drifts, the engine silently measures something other than what the file says.

`NOTES.md` ends with a section titled **Not verified**, and it must not be empty. This is the single highest-return convention in the pipeline. One package listed four properties it could not establish; three turned out to be genuine issues and the fourth was resolved in minutes — because it had been named. Naming what you do not know is not weakness in a research report. It is most of the value.

Identity

Two strategies must never be confusable, across time, across researchers, across machines. So an identifier is minted from **content**, not from a name.

```
W3 - OD - MNQ - BACRBD
|   |   |   |
|   |   |   | checksum, derived from the content
|   |   |   | instrument
|   |   |   | family, from a shared vocabulary
|   |   |   | originating workspace
```

The checksum is the first 30 bits of

```
sha256( workspace | family | symbol | strategy_hash | params_hash )
```

rendered in Crockford base32, where:

- strategy_hash** is the SHA-256 of the *normalised* source — line endings unified, trailing whitespace stripped, blank lines and comment-only lines dropped — truncated to 32 hex characters. Reformatting does not move it; changing a line of logic does. The identity follows the logic, not the formatting.
- params_hash** is the SHA-256 of a canonical JSON of the parameter class name and its sorted values, floats rounded to fixed precision, truncated to 16 hex characters.

Four consequences, and they are the point of the design:

1. **Identical content mints an identical identifier.** Re-exporting an unchanged strategy does not create a new identity, because it is not a new strategy.
2. **Any change to the logic mints a different one.**
3. **A changed parameter mints a different one.** The same file at two different risk settings is two strategies with two different results and two identities. This is not pedantry: a firm that calls them by one name eventually cannot say which configuration produced which number.
4. **The parameter class name is inside the hash.** Renaming the class moves the identity even with identical values, which is intended — it is a different declared object.

The ledger

Every minted identifier is appended to a ledger that is never edited or reordered: identifier, timestamp, strategy, both hashes, symbol, family, source path, and a note. Minting refuses if the identifier already exists with different content, which turns a one-in-a-billion silent collision into a loud failure at export time.

The ledger is also the **honest trial count**. Statistical corrections for multiple testing need to know how many things were tried, and a register that only records successes gives an answer that is wrong in the direction that flatters you. Screens rejected before a source ever existed — the twenty-plus families of earlier batches, the 1,287 cells of one sweep — have no ledger row, so the count is handed over together with the research ledger that does record them.

Worked example

The strategy in section 9 has three identities from one file:

Configuration	Identity
class defaults (reference only, never traded)	W3-OD-MNQ-BACRBD
evaluation profile: looser gate, larger stake, wider target	W3-OD-MNQ-0ZYC0A
funded profile: smaller stake, volatility filter on	W3-OD-MNQ-BC1TX8

One source, one `strategy_hash`, three `params_hash` values, three identities. The two traded profiles are recorded as explicit overrides in the route configuration, so what was measured and what would be deployed are the same object.

Packaging discipline

Three rules, each learned expensively:

Freeze then mint, never the reverse. The identity is a function of the frozen content. If a package is edited after minting, it must be re-frozen and re-minted — there is no such thing as a small edit to a shipped package.

Never repair a frozen source to make a check pass. During the export in section 9, an automated gate found a genuine unit bug in a package that had already shipped and was awaiting measurement. The source was **not** edited. The defect was written into a known-defects register with the mechanism, the consequence and a recommendation, and the decision to re-freeze was left to the owner. Silently fixing it would have destroyed the link between the package and the evidence measured from it.

State dependencies; do not quietly design around them. The same export initially had a feature removed because the destination system appeared unable to support the data feed it needed. That was the wrong call — it discarded the single measured improvement of an entire research programme to satisfy a constraint nobody had confirmed. The correct move, and what the package now does, is to keep the feature, declare the dependency loudly, ship the feed specification and the data table so implementing it is a small job, and offer an inert fallback setting.

8. Worked example one: the idea that died

Read this section closely. It is the complete arc of a research family, from a practitioner claim to a measured verdict, in two days. It is the most representative thing in the pack.

The idea

Options dealers who sell options carry gamma risk and must delta-hedge continuously. When their book is net **short** gamma they hedge in the direction of the move — buying as price rises, selling as it falls — which amplifies it. When net **long** gamma they hedge against the move, damping it and pinning price near strikes with large open interest.

Practitioner writing goes further: the level where net dealer gamma crosses zero, the *flip*, is supposedly a regime boundary, with trending behaviour below it and mean-reverting behaviour above.

The mechanism is genuinely attractive by our own filter in section 2. There is a named payer — the hedger paying for immediacy — the flow is mechanical and price-insensitive, and its *location* is computable in advance from public data. That is more than most ideas offer.

It was also, at the time, the most-hyped idea in retail and semi-professional market commentary, which is neither evidence for nor against it. This firm had already established something adjacent and unwelcome: implied volatility predicts the *magnitude* of a session's move and says nothing about its *direction*. That prior mattered, and it turned out to be the whole story.

Getting the data

The firm held no option data. The hypothesis was framed, an options subscription was purchased the same day, and the full six-year history was downloading within the hour: end-of-day greeks, implied volatility and open interest by strike and expiry.

The download took an hour and produced 6,164 files under a hash manifest. Before any hypothesis touched it, the data was checked against what the vendor claimed — which is where the first real finding appeared.

The publication times were not what the documentation said. A causal assertion in the feature builder — "no row may be used before the moment it was knowable" — refused two entire years of data. Investigation showed the vendor batch-stamps early-era reports at 07:01 rather than the documented 06:30, and emits late tails as far as 20:15 in every era. The fix was row-level enforcement of the actual report time, with the excluded mass recorded per session; not a looser clock.

That assertion was ten lines of code. Without it, two years of look-ahead would have entered the study silently and every result afterwards would have been worthless in a way that is nearly impossible to detect from the outside.

The model

Section 3.3 has the mathematics; `code/gex_math.py` runs it. Per session, using only information knowable before the opening auction:

1. take open interest by strike and expiry, published overnight, describing the prior close;

2. take greeks and implied volatility from the prior session's end-of-day report;
3. compute each contract's dollar gamma per 1% move, $\Gamma \cdot 0I \cdot m \cdot S^2 \cdot 0.01$;
4. sign each line by the dealer convention and sum to get net gamma $N(S)$;
5. find the flip level as the zero crossing of N nearest spot;
6. map it onto the futures tick grid through a **measured** basis.

Every session also carries a quality figure — the share of the option book, weighted by open interest, that arrived on time and could be priced — and sessions below 85% were excluded before any outcome was opened.

First death: no events

The preregistered primary endpoint was the 30-minute continuation after price first *touches* the flip level, compared between short-gamma and long-gamma sessions, with a floor of 100 events per group.

The count came back at **52 and 70**.

An epsilon ladder — widening the touch tolerance from 2 ticks to 12 — moved the short-gamma count from 52 to 58. The tolerance was not the limiter. Price simply approaches the flip at all on only about a quarter of sessions, and no honest definition of "touch" reaches the floor. Extending the window into the calibration partition reached 82, still short, and would have spent a sealed partition to get there.

Dead at the sample gate, with zero outcome statistics ever computed.

That last clause is what made the next step legitimate. Because the dry run consumed only *counts*, no outcome existed to be fitted to, and the redesign that followed was a genuine preregistration rather than a response to a disappointing result. Had the counts been computed alongside the outcomes — which would have been more convenient — the redesign would have been post-hoc tuning wearing a protocol's clothes.

The lesson, and it is cheap: check the event rate before you freeze an endpoint. Counting is free. Freezing an endpoint whose sample cannot exist is not.

The redesign

An amendment moved the measurement from the rare touch to the mechanism's natural scale: every qualifying session becomes one observation.

For each session, with d_0 the displacement from the opening reference to the flip level and m the session's move:

$$A = \text{sign}(d_0) \cdot m$$

A is positive when the session moved *away* from the flip. The primary endpoint became the difference in mean A between short-gamma and long-gamma sessions. The mechanism predicts it is positive. One observation per session, 243 and 211 sessions — the sample gate now cleared with margin.

Everything else was carried over unchanged: partitions, quality gate, controls, gate order.

Second death: no effect

Primary differential	−7.43 ticks
95% session-bootstrap interval	[−31.75, +18.93]
Real level minus phantom levels	−5.16, interval [−19.70, +10.55]
Inverted-convention control	exact arithmetic negation, mismatch 0.0
Random-sign control	not significant, as a true null requires
Era split	signs disagree across the structural break

Zero inside the interval, and the sign is the opposite of the prediction. The phantom control is the more informative line: **the real flip level behaved no differently from a level 25 to 50 points away.**

Gate 2 failed, so gates 3 through 6 were **not evaluated**. No rescue diagnostics were run.

Two strategies had been written against this hypothesis. Both died **without ever receiving a backtest**, which is the cheapest death a strategy can have and exactly what the preregistration existed to produce.

Confirming the null three more ways

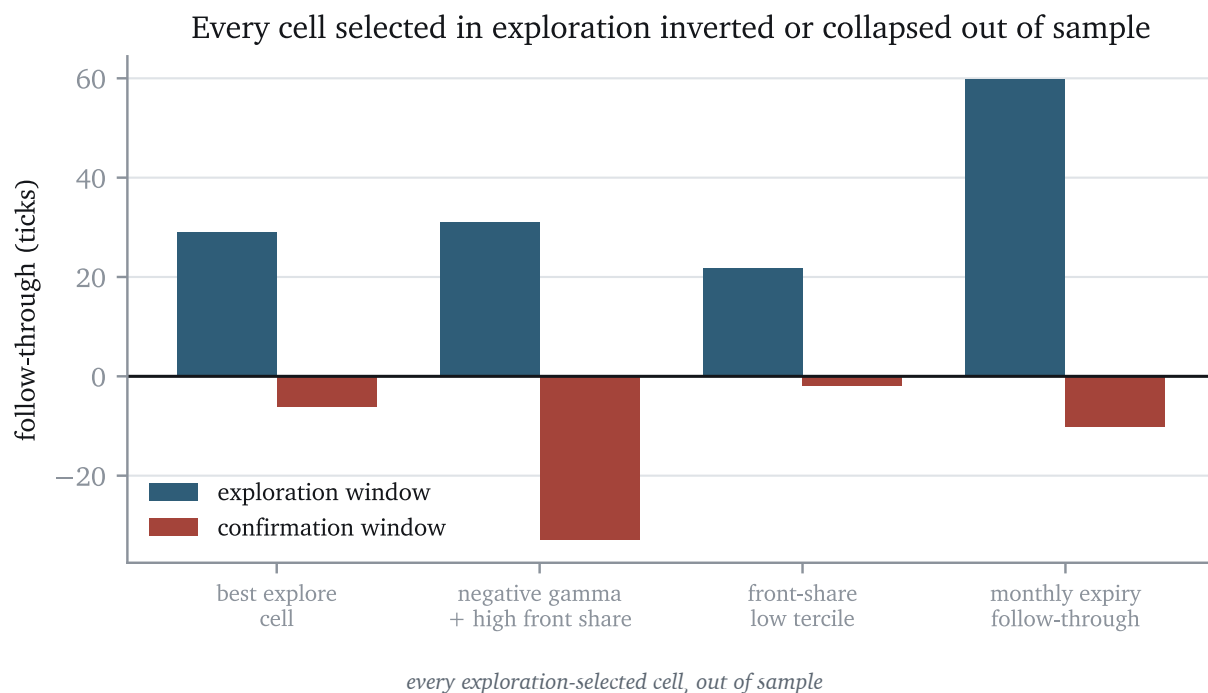
With the primary endpoint dead, three independent exploratory lanes ran in parallel over roughly 1,400 cells, on the last five years, with an explore/confirm split reported side by side.

Lane A — gamma regime conditioning an opening-range breakout. Every cell selected in exploration inverted or collapsed in confirmation. The best went from +29.0 ticks at $t = 2.6$ to −6.1. The theory-aligned cell — negative gamma with a high share of near-dated gamma — went from +31.1 at $t = 2.3$ to −32.9 at $t = -3.0$. The practitioners' claim did not merely fail; in the confirmation window it inverted.

Lane B — expiry and pinning, 1,287 cells. Zero cells positive in both windows. The apparent "convergence toward large strikes" turned out to be a diffusion artifact: under a random walk the distance to a fixed nearby point *increases* in expectation, so the effect was present in the null as well. Monthly-expiry follow-through flipped sign between windows, +59.8 at $t = 3.5$ to −10.1.

Lane C — the volatility surface as a directional signal, 54 cells. Best exploration t-statistic: 1.62. Independent noise alone would be expected to reach 2.13. The search did not clear its own null.

Finally, the underlying breakout was measured through the engine with real fills: **968 trades, −\$20,831, win rate 31.5%** — which is the geometric null for that bracket. The base strategy had no edge either, so the conditioning had nothing to condition.



What survived

One thing, and it is not directional.

Trade days with **high near-dated implied volatility** carried **-\$718** per trade at a 10% win rate in the confirmation window, against **+\$629** and 48% on low-volatility days. Used as a filter on a funded account, it cut account failures from **9 of 13 to 2 of 13** with payout eligibility unchanged.

Used on an *evaluation* account it was actively harmful: filtering removes trade days, which lengthens a fee-paying attempt from a 15-session median to 79 and nearly quadruples the fee per pass. Survival is what a funded account buys; speed is what an evaluation buys, and the same filter is right for one and wrong for the other.

The general result, now confirmed three independent times at this firm: the options book prices magnitude and survival, never direction. Ask it about size and risk. Do not ask it which way.

The register entry

The family is closed with what was tested, what closed it, and what would make a retest legitimate. Stating the boundary precisely is the difference between a useful register and a superstition:

What is falsified is **direction**, from the **overnight-carried** book, at **session** scale, on the **S&P** contract. Intraday-updated positioning is a different object and is untested. Nasdaq-native features are untested. Volatility-regime stratification of evidence is untested and is an infrastructure use rather than a signal.

What this cost, and what it bought

Two days, one eighty-dollar subscription, and a research pipeline that is now permanent infrastructure. It bought a definitive answer to a question the whole retail market is currently arguing about, one measured

filter that is in production, three reusable lessons, and a register entry that will stop the next person spending a month on it.

That is a good outcome, and it is what most of the job looks like.

9. Worked example two: the thing that shipped

The dealer-gamma family died on a Sunday afternoon. A deliverable shipped the same day — from a different direction entirely, and the contrast is the lesson.

Where a shipped strategy usually comes from

Not from a new mechanism. From the best mechanism you already have, improved along an axis nobody has measured.

The firm's most reliable family is **opening-drive continuation**: a large overnight displacement, a same-direction drive in the first half hour of the cash session, and a confirmed breakout with trend, momentum and order-flow agreement. One member of that family had already been measured at tick level by the firm's engine and runs on a live demo account.

The question asked was not "what else could work" but: **the account pays monthly rent, so what happens if we optimise for speed instead of reliability?**

The reasoning, before any code

The existing strategy passes an evaluation in a **52-session median**. Every month of that costs a subscription. Two levers are available, and both trade reliability for speed:

- **loosen the entry gate** — more qualifying sessions, so the first opportunity arrives sooner;
- **increase the stake** — fewer trades needed to clear the target, so an attempt resolves faster.

Both make the account *more* likely to fail. Conventional risk management says that is wrong. The economics say otherwise, and the firm had already measured it: reducing risk per trade on an evaluation strategy raised the pass rate from 42.9% to 66.7% and dropped net profit from **+\$13,989** to **+\$148**, because smaller size meant twice as many trades per attempt, each attempt ran longer, and each accumulated more rent.

For an account that pays rent by the month, time is a cost, and the correct direction for an evaluation lane is more risk per trade, not less. That is a counterintuitive, measured, and narrow result — it applies to the evaluation stage only. The funded stage, where survival is the objective, keeps the opposite rule.

So the change was a **constant-transform** of a tick-verified strategy: identical logic, identical code path, three constants moved. Nothing about the signal was touched, which means the tick-level evidence for the mechanism still applies and only the risk geometry is new.

The measurement gap

To answer the question, we needed a statistic nobody had been computing: not "did this account survive" but **"how many evaluations does this pass per year, and what did the attempts cost"**.

So two instruments were built before any parameter was tried:

A cadence harness. Non-overlapping single-account runs on a rolling grid of start dates. Each run reports its terminal state, days elapsed, trades, and an estimated fee including monthly renewals. The aggregate gives pass rate, median days to pass, and fee per pass.

A payout probe. For the funded stage, the published rule implemented directly: a non-withdrawable buffer plus a minimum request, no single day above half the cumulative profit, and a minimum number of profitable days. It reports what share of funded accounts reach a first withdrawal, and how many die first.

Building the measuring instrument before running the experiment is not overhead — it is the experiment. And it paid immediately: the harness surfaced a defect in the research engine's own risk layer, where a conservative buffer was converting account failures into a limbo state in which nothing traded and rent accrued indefinitely. Switching it off changed the terminal-state distribution completely and made the measurement match live behaviour.

The result

Twenty-five non-overlapping starts across five years, stress cost assumptions, pooled with an independent grid offset by 45 sessions:

	Existing strategy	Speed-optimised profile
Pass rate	30.8%	24.0%
Median sessions to pass	52	15
Fees per pass	~\$1,492	~\$759
Mean profit on a pass	~\$4,308	~\$5,580

A lower pass rate, and **half the time and half the cost per pass**. That is the trade the economics predicted, confirmed on data that had not been used to design it.

Two supporting numbers mattered as much as the headline:

Concentration. On an unconstrained account so that no early termination truncates the sample: 143 trades, mean **+\$500**, and a **top-5%-trimmed mean of +\$188**. The result survives removing its best trades, which is the test that has killed more candidates at this firm than bad signals have. Five of six years positive.

Plan structure. The same profile on the other account plan: **1 pass in 13 starts** at nearly four times the fee. That plan caps any single day at half the total profit and requires two trading days, which structurally forbids the one-trade pass this geometry produces. The account plan is a research variable, not a deployment detail.

The credibility check

The harness is new, so before trusting it, it was pointed at the existing strategy whose behaviour the firm's engine had already measured independently.

Local pass rate: **30.8%**. The engine's out-of-sample number: **30.8%**.

That agreement is what earns the new numbers the right to be read at all. **A new measuring instrument gets calibrated against a known answer before it is used to produce an unknown one.**

Packaging, and a defect found on the way out

The package shipped as one source with two profiles — a fast evaluation profile and a survival-tuned funded profile carrying the volatility filter from section 8 — each with its own identity, and a specification and data table for the feed the funded profile needs.

An automated packaging gate was written for the export: it recomputes every hash, checks the parameter file against the class defaults, recomputes each profile's identity from its declared overrides, and cross-checks the index and route configuration against the manifests.

Its first run found a genuine unit bug in a package that had already shipped and was waiting to be measured. That strategy stores a reference level as a tick index and compares it against a display price. On the Nasdaq contract the units differ by a factor of four, so the computed displacement is around $-63,000$ on every session regardless of what the market did, which pins the direction to short on every trade, renders a magnitude threshold permanently inert, and kills a cancellation path entirely. Every performance figure attached to it describes a different object than the one its documentation describes.

The frozen source was **not** edited. The defect was written into a known-defects register with the mechanism, the consequence and a recommendation, and the re-freeze decision left to the owner.

That is the third instance of the same unit error in this firm's history, which is why section 5 puts branch reachability counts ahead of everything else, and why the check is now mechanical.

What to take from the contrast

Section 8 is a good idea, well tested, that turned out to be false. This section is an existing mechanism, re-optimised along an axis the economics pointed at, that turned out to be worth shipping.

Both were the same day's work and both are the job. The difference between them is not skill or luck — it is that one asked "is this true?" and the other asked "given what is already true, what follows?". The second question has a much higher hit rate, and a researcher who only asks the first will produce a lot of well-documented nothing.

Ask both. Spend most of your time on the second.

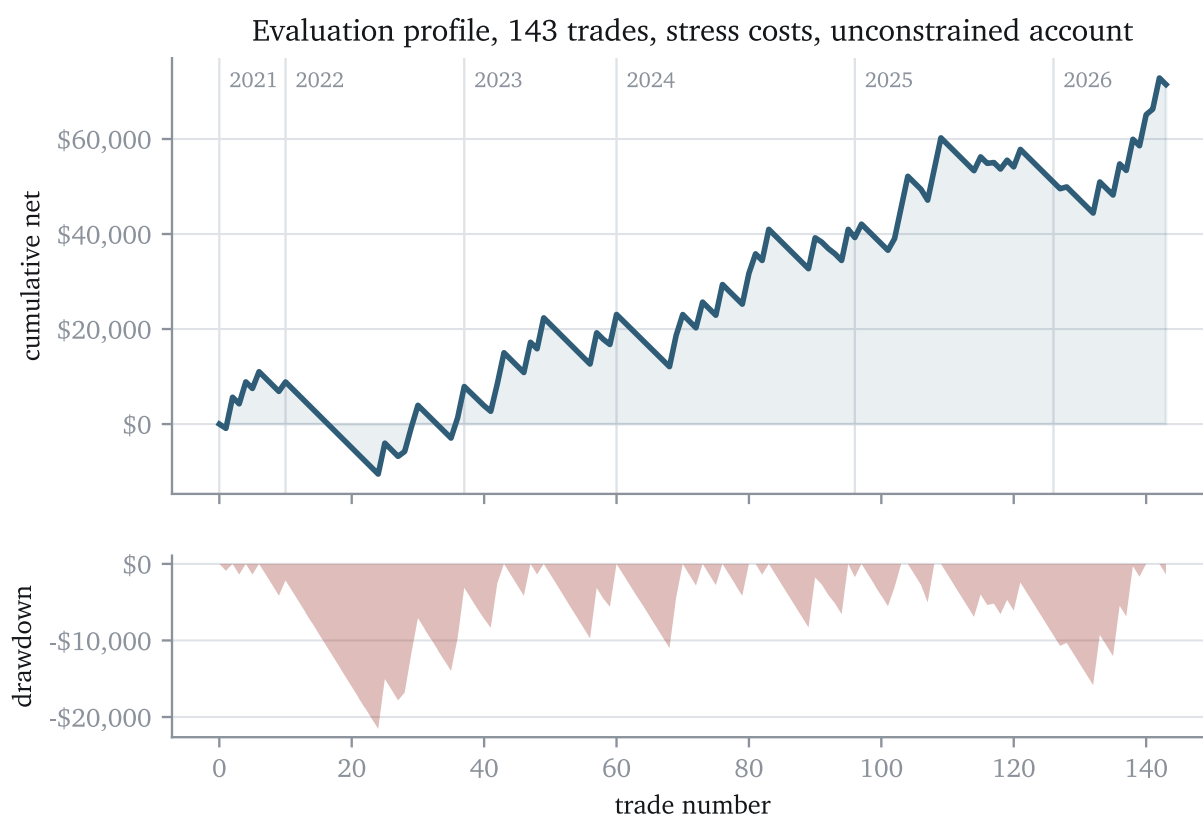
10. Preliminary results

Everything on this page is **screening evidence**, produced by the research engine on one-minute bars. It has not been through the firm's tick-level measurement, and until it has, none of it is a verdict. Read section 1 if you skipped the distinction; it matters most exactly here, where the numbers look good.

The strategy is the one described in section 9: the opening-drive continuation mechanism at a speed-optimised risk setting, on the Nasdaq micro contract, 2021-08 to 2026-08, with stress cost assumptions throughout.

The trade record

143 trades over five years on an unconstrained account — that is, an account with no drawdown limit and no profit target, so that nothing truncates the sample and every trade the strategy would have taken is present.



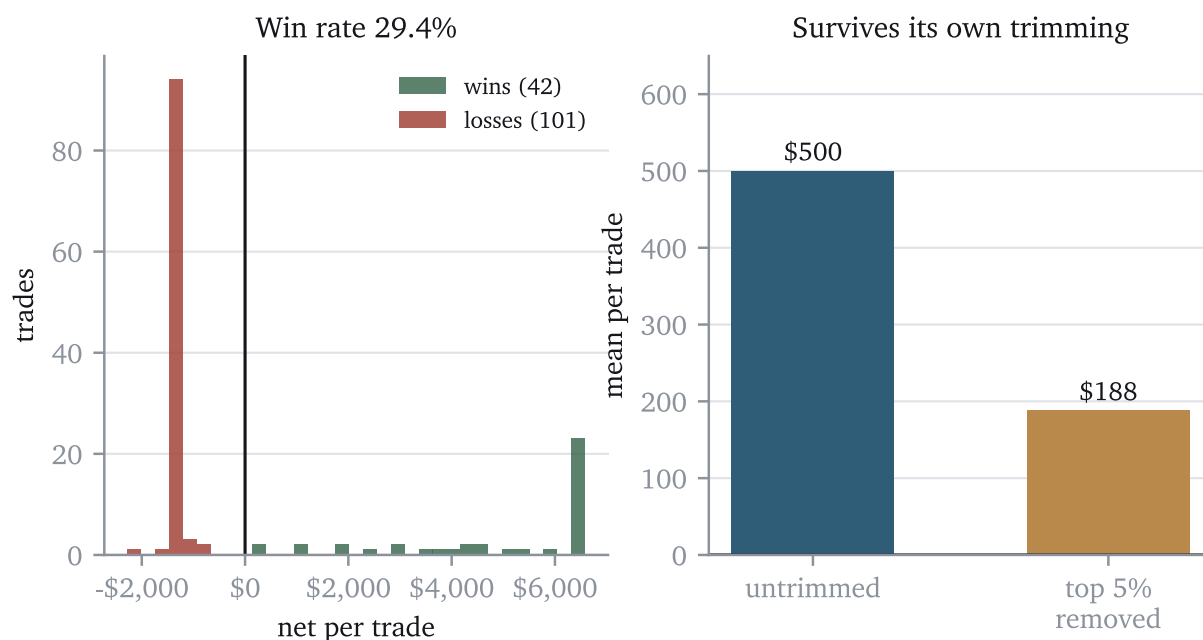
cumulative net and drawdown across the full trade record

The shape is what an asymmetric continuation strategy is supposed to look like: long flat stretches punctuated by large winners, with drawdowns that are shallow relative to the eventual gains because the losses are capped by design.

Trades	143
Net, stress costs	+\$71,467
Mean per trade	+\$500
Top-5%-trimmed mean	+\$188
Win rate	29.4%
Best / worst trade	+\$6,585 / -\$2,282

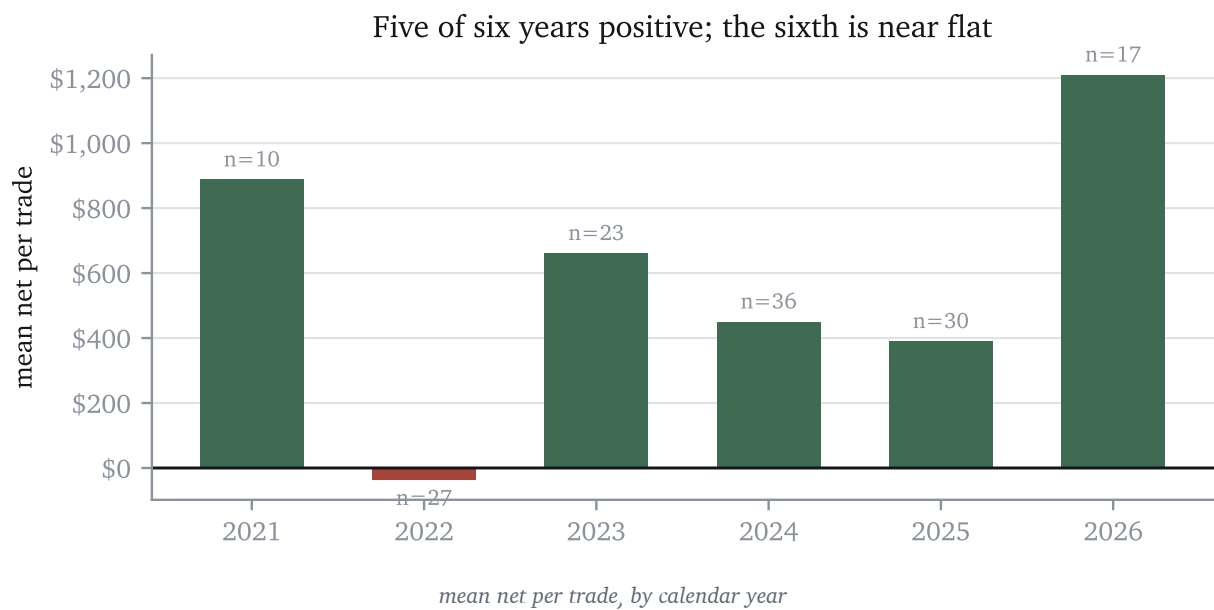
Is it a handful of lucky trades?

This is the first question to ask of any result with a 29% win rate, and the firm has killed more candidates on this test than on any other.



the distribution of trade outcomes, and what survives trimming

Removing the best 5% of trades leaves the mean at **+\$188**. The result does not depend on a handful of outliers. Note also the shape of the loss side: losses are tightly clustered because the stop is a fixed distance, while the win side has a long right tail. That asymmetry *is* the strategy — it is not a defect to be engineered away.

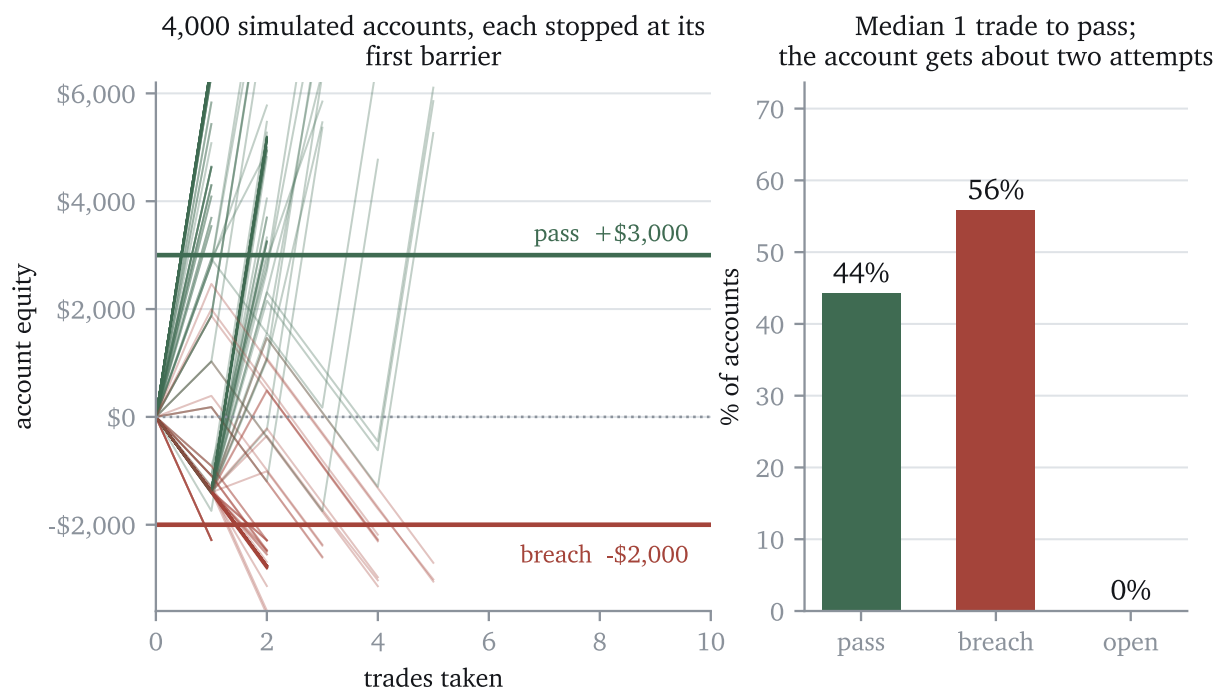


Five of six years positive, the sixth close to flat, and no single year carrying the result.

What happens to an actual account

The numbers above describe the *signal*. An evaluation account is a different object: it stops the moment it touches either barrier, and that truncation changes everything about the outcome distribution.

So we resample the real trade record into four thousand simulated accounts, each starting fresh and each stopped at its first barrier — pass at +\$3,000, breach at −\$2,000.



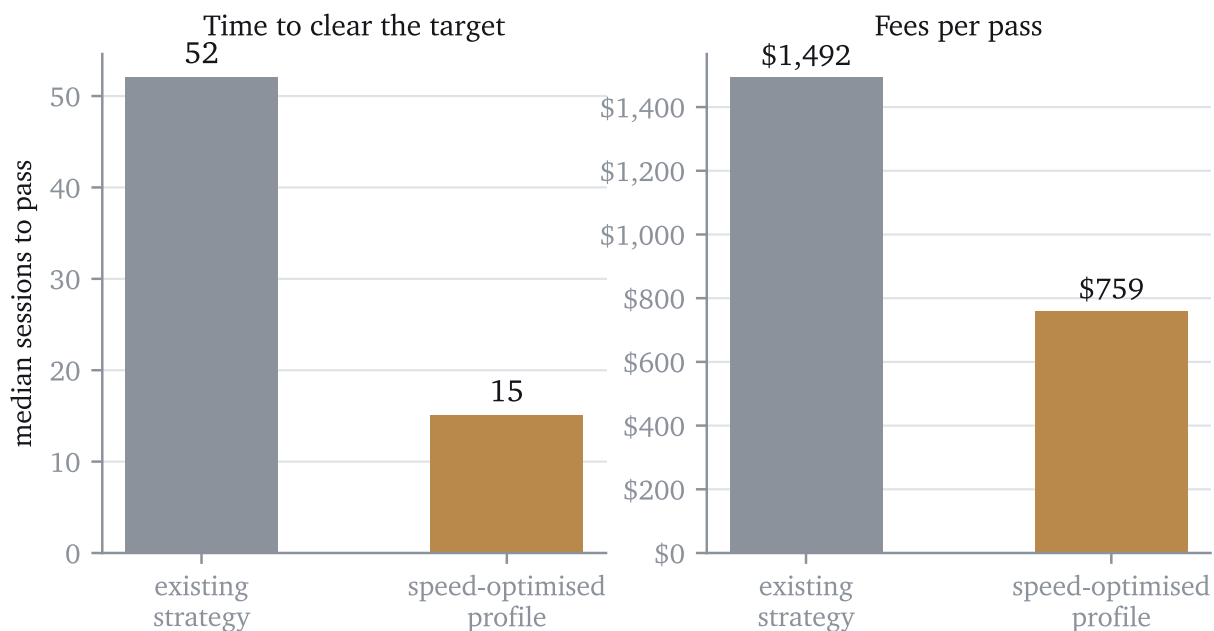
Accounts that pass	44%
Accounts that breach	56%
Median trades to a pass	1

That median of one trade is not a rounding artifact. At this risk setting a single winning trade clears the entire profit target, and two consecutive losses exhaust the drawdown allowance. **The account is a two-shot bet**, and its pass rate is close to the strategy's win rate because there is very little inside a two-trade account for risk management to manage.

That is a legitimate shape for a challenge lane, and it must never be described as a low-risk one.

Why this beats a more reliable strategy

The Monte Carlo above assumes a fresh account. The economics that matter run over many accounts, because the subscription is paid monthly for as long as each attempt takes.



time to pass and fees per pass, against the existing strategy

Measured over twenty-five non-overlapping single-account starts, pooled with an independent grid offset by 45 sessions:

	Existing strategy	Speed-optimised profile
Pass rate	30.8%	24.0%
Median sessions to pass	52	15
Fees per pass	~\$1,492	~\$759
Mean profit on a pass	~\$4,308	~\$5,580

A lower pass rate, and half the time and half the cost per pass. For an account paying rent by the month, that trade is the right way round — and it is the opposite of what conventional risk management would advise,

which is why it had to be measured rather than argued.

The credibility check

A new measuring instrument gets calibrated against a known answer before it is used to produce an unknown one.

The cadence harness that produced the table above was pointed at the existing strategy, whose behaviour the firm's tick-level engine had already measured independently. Local pass rate: **30.8%**. The engine's out-of-sample figure: **30.8%**.

That agreement is what earns the rest of these numbers the right to be read.

What these numbers are not

Stated plainly, because a page of favourable results is exactly where a reader stops being sceptical:

1. **No tick-level verdict exists.** These are one-minute-bar fills. The research engine arms protective bracket orders one bar after entry, so intrabar stop-and-target ordering is unresolved.
2. **Twenty-five account starts is a cadence estimate, not a distribution.** The confidence interval on a 24% pass rate from 25 starts is wide, and we have not pretended otherwise.
3. **The risk constants were chosen on this data.** The underlying signal is inherited unchanged from a strategy that has been measured at tick level; the three constants that make this profile fast were selected here, and there is no untouched holdout behind that choice.
4. **The Monte Carlo resamples the observed trades independently.** It therefore assumes trades are exchangeable and independent, which discards any serial structure — clustering of wins in trending regimes, for instance. It is a distributional sketch, not a forecast.
5. **Six years of history contains no crisis regime.** The data begins in August 2020.
6. **Nothing here models the funded stage,** so every figure on this page describes what passing *costs*, never what passing *earns*.

A candidate who reads this page and asks about item 3 or item 4 unprompted is the kind of candidate we are looking for.

11. What gets work thrown out

Every item is a real failure from this firm's history. None is a style preference. Read this before your first strategy and again before your first export.

Failures that make a strategy unmeasurable

These are the worst kind, because the strategy still runs, still reports, and still produces a verdict — about something other than what you think.

Mixing price units

Every price the engine hands a strategy is an **integer tick index**, not a price. On the Nasdaq contract they differ by a factor of four. Comparing one against the other does not raise; it produces a number.

Measured on one real strategy, over the 32,444 bars where every input was finite:

condition	before the fix	after
close > long level	0	11,124
close < short level	32,444	8,456

The long branch was unsatisfiable and the short condition was true on every bar, which made its crossing requirement unsatisfiable in turn. **The whole branch was dead in both directions** — not biased, dead — and the strategy had produced a full set of results.

This has now happened **three times** at this firm, most recently in a package that had already shipped. The defence is mechanical: publish the series you intend to compare and read it back by name, then count how many bars satisfy each clause.

Sizing off a quantity that ratchets

Live headroom — the distance to your drawdown floor — is not a risk budget. Under a trailing drawdown the floor rises with your high-water mark and **never retreats**, so after the first drawdown headroom stops being a budget and becomes a distance that only shrinks.

Two strategies sized off it. Headroom froze at \$506.92, the computed quantity floored to zero, and the strategy was **silent for 2.25 million bars** — nine trades in seven years — while reporting a healthy status the entire time. Nothing breached, because nothing traded.

Latch a base budget once from the account's *starting* allowance. Use live headroom only as a cap that can shrink the position, never as the base.

A latch with no clearing condition

Two strategies set a portfolio lock when drawdown reached a threshold and cleared it **nowhere**. The lock closed 25 sessions into a seven-year backtest and stayed closed for the remaining 6.9 years.

Every state variable that survives a day needs a documented clearing condition and a test for it. Initialisation at start-up is not a clearing condition. If the intended condition genuinely is "never", say so in the notes and accept that the backtest measures only the period before it fired.

`quantity == 0` as a silent return

A strategy that declines to size is indistinguishable in the output from one that saw no signal. That is precisely how the failure above went unnoticed for an entire research programme. Record it as an event.

Failures that make evidence meaningless

Windowed grids instead of a contiguous run

The tempting design is a grid of overlapping fixed-length windows, each starting a fresh account. It is a start-date sensitivity study, and it is **structurally blind to any state that survives a day** — a latch, a counter, a peak, an account floor.

Two strategies scored five of eight joint passes on such a grid. The same strategies in a contiguous single-account run traded almost nothing for years. Both reports were arithmetically correct about different experiments; only one was about the strategy.

Reporting a mean without its concentration

Report the mean after removing the top 5% of trades beside the untrimmed mean, and a per-year panel.

One strategy earned `+17.68` ticks per trade. Twenty trades out of 616 — three trading days a year — supplied **55% of everything it earned in seven years**. Remove them and it lands within 4% of the threshold it was supposed to clear.

Concentration is not automatically disqualifying; breakout strategies are supposed to look like this. What is disqualifying is reporting the headline without it.

A win rate without its geometric null

A win rate is a statement about barrier geometry until you prove otherwise. In a sweep of twenty mean-reversion cells, widening the stop and shrinking the target moved the win rate from 13.9% to **79.3%** while the mean stayed pinned near zero. That monotone gradient with a flat mean is the signature of no drift — it is what a martingale looks like.

Report your win rate next to `1/(1+RR)` every time.

Optimising against a subsample selected by a filter

An early study recorded the research engine as "3.5× more pessimistic" than a hand reconstruction, which invited a correction factor. The gap was an artifact of comparing against a gate-truncated subsample rather than a random one. On like-for-like samples the two agreed within noise.

When two measurements disagree, first prove they are being fed the same trades. Almost every cross-engine discrepancy in this firm's history has been a comparison error.

Failures of process

Editing a frozen artifact to make a check pass. Frozen sources, protocols, manifests and hash constants are an audit record, not configuration. If an upstream change breaks a frozen check, write a disclosure and let the owner decide. Making the suite green by rewriting what was known at decision time destroys the only thing that made the record worth keeping.

Running diagnostics after a gate has failed. Once a kill gate fails, every further statistic is a search for a reason to continue. Stop.

Reporting the exploration window and calling the confirmation window "needs more data". A cell that inverts out of sample is a negative result. Say so.

Narrowing a definition after seeing outcomes. Before outcomes, it is modelling. After, it is fitting. The distinction is invisible in the final artifact unless the protocol was frozen first, which is the entire reason protocols are frozen.

Quietly designing around an assumed constraint. If a downstream system appears unable to support something your work needs, state the dependency and ship whatever makes it cheap to implement. Do not silently remove the feature — the constraint may not be real, and you will have thrown away a result to satisfy it.

12. How the work actually runs

The rhythm

There is no ticket queue. A researcher owns a hypothesis from the literature search to the ledger entry, and the unit of work is a **batch**: one family, one protocol, one verdict, one record.

A batch is typically one to three weeks. The dealer-gamma family in section 8 was unusually fast at two days, because the sample gate killed the first endpoint before any expensive computation ran — which is the pre-flight discipline paying for itself.

What you are expected to produce

Whatever the outcome, a batch ends with the same five artifacts. This is the deliverable, and a negative result produces all five exactly as a positive one does:

A frozen protocol	with its hash in the ledger, written before outcomes were opened
Canonical evidence	the actual numbers, versioned, tied to a hashed input
A ledger entry	append-only: what was decided, on what evidence, on what date
A register entry	if the hypothesis is closed: what was tested, what closed it, what would make a retest legitimate
A durable lesson	if the batch produced one that generalises beyond its family

What good looks like

The researcher who is valuable here is not the one with the most ideas. It is the one whose negative results are trustworthy enough that nobody has to repeat them.

Concretely:

- **Kills their own ideas fast.** The pre-flight arithmetic is ten minutes and it should be the first thing you reach for, including — especially — when you like the idea.
- **Writes the falsifier down first.** If you cannot state in advance what observation would make you abandon the hypothesis, you are not going to abandon it.
- **Reports the number that closed it.** "It didn't work" is not a result. "−7.43 ticks, 95% interval [−31.75, +18.93], and the real level was indistinguishable from a phantom 100 ticks away" is a result, and it is permanent.
- **Builds the measuring instrument before the experiment,** and calibrates it against a known answer first.
- **Names what they could not verify.** Every report ends with that section and it is never empty.

Where the work lands

Everything you research is aimed at a **\$50,000 prop-firm evaluation account** — see section 1. You are not researching for a hypothetical book, and you are not researching for the firm's own capital on day one. The

route is concrete:

your research (this pack)	->	frozen package (hashed)	->	tick-level measurement (the verdict)	->	live demo account (real, prop-funded)
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A candidate that survives all four stages is funded on a prop evaluation account and traded for real. The firm carries the subscription cost; you carry the research. Researchers whose strategies survive that route take on more over time, including the firm's own capital — but the prop route is where the track record is built, and it is where everyone starts.

Two consequences worth internalising early:

- **Your objective function is the account's rules**, not an abstract Sharpe ratio. Time-to-pass, fees, and the drawdown floor are first-class research variables, and section 9 shows a case where optimising for them reversed the conventional answer.
- **A failed account is a bounded, expected cost**, not a disaster. The aggressive lane accepts that many accounts fail; what it does not accept is not knowing why.

What we provide

- **Data.** Everything in section 4, and anything else the work needs — see the acquisition policy there. It is not a formality; it has been exercised.
- **Compute.** Local, and sized so that a full six-year backtest is minutes rather than hours.
- **A measurement engine you do not have to build.** Tick-level fills against real quotes under the firm's risk model, with a placebo suite, a Monte Carlo, walk-forward, and a look-ahead detector. Your job is to produce candidates worth its time.
- **The record.** Every result the firm has ever produced, including every failure, with the number that closed it. You will spend real time reading it, and it will save you more.

What we ask

Speak plainly about weak evidence. The failure mode this firm guards against hardest is not a bad idea — it is a mediocre result described in language that makes it sound better than it is. There is no penalty here for saying "this does not work", and there is a large one for saying "this looks promising" about something that does not.

The exercise

`exercise/README.md` contains a task with a self-contained dataset. It takes a few hours and it is the fastest way for both sides to find out whether this work suits you.

We are not looking for a strategy that makes money. We are looking at how you decide whether something is real.